Caleb Woldemedhine

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EDUCATION

University of Denver Denver, CO
B.S. in Computer Science (AI & ML Concentration) Sept 2023 - Jun 2027
Minors: Mathematics and Statistics
GPA: 3.66
Relevant Coursework: Database Organization & Management, Topics in Artificial Intelligence (Machine Learning & LLMs), Applied Math for Data Science and AI, Machine Learning for IoT and Embedded Devices, Linear Algebra, Introduction to Probability

EXPERIENCE

Cincinnati Children's Hospital Medical Center Remote
Machine Learning Research Intern — Prasath Lab Aug 2026–Present
- Developing and evaluating machine-learning and agentic AI methods for digital pathology, using LLMs and vision-language models to support whole-slide image analysis and diagnostic reasoning.
- Conducting a technical review of recent pathology-agent systems, comparing their architectures, datasets, tool-use workflows, evaluation metrics, and clinical limitations to guide model development.

Palo Alto Networks Santa Clara, CA
Software Engineering Intern June 2026 - Present
- Built an LLM visualization service on a Pub/Sub backend that profiles datasets, selects the correct chart, and converts agent outputs into frontend JSON contracts. Created a sub-agent that infers data schemas and generates mock datasets to test 16 chart types and edge cases, achieving 94% chart-selection accuracy while ensuring all outputs could render in the UI.
- Refactored the system into a single-spec chart catalog and migrated the LLM engine to Vertex AI. Built a seven-worker concurrent E2E testing system that automated multi-step Pub/Sub and HTTP flows, along with BigQuery tables and external views used for schema grounding, test validation, and monitoring.

Frederick S. Pardee Institute for International Futures Denver, CO
Data Technician Sept 2025 - Aug 2026
- Cleaned and preprocessed international education datasets using Python and pandas, standardizing country-year coverage, resolving missing or inconsistent values, and running quality checks to prepare tables for database imports and website updates.
- Built a Python import script to convert cleaned Excel datasets into SQL-ready inserts, automate formatting and loading workflows, and verify imported values against the Data Dictionary for accurate database integration.

Tempus Automation Highlands Ranch, CO
Software Engineering Intern Jun 2025 - Aug 2025
- Built and evaluated a C# ML.NET model using a custom dataset of computer-vision-derived deck dimensions, material specifications, pricing data, and design constraints to predict material quantities and total project costs, then compared its predictions against expected results from test records.
- Designed a MySQL database for inventory, pricing, and training data, then integrated the ML.NET model into a C# backend that queried the database and sent material recommendations and cost estimates to the frontend as JSON responses.

Denver Center of African Art Denver, CO
Web Development Intern Jun 2024 - Sept 2024
- Built a responsive website using HTML, CSS, and JavaScript and integrated Stripe donation payments with receipts and donor record tracking.

PROJECTS

Stock Prediction and Analysis Tool
- Built a stock analysis and forecasting website with a Python backend and JavaScript frontend that gives users an interactive dashboard with price forecasts, sentiment insights, trading signals, and model accuracy results powered by machine learning.

DuDraw AI Coder
- Built a full-stack AI website where users enter a prompt and a large language model generates DuDraw code that turns their ideas into interactive drawings and simple games.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, C, C++, C#, SQL, JavaScript, TypeScript, HTML, CSS, React, Next.js, R, Haskell
Libraries & Technologies: PyTorch, TensorFlow, Keras, Scikit-learn, Prophet, Pandas, Plotly, NumPy, Excel, Tableau, REST APIs, AWS, ML.NET, Docker, Git, Linux, JSON

LEADERSHIP

National Society of Black Engineers Denver, CO
President Jun 2025 - Present
- Raised $40,000 to send 35 students to the NSBE National Convention and organized a multi-company hiring fair, leading employer outreach, sponsorships, and event operations to expand student career opportunities.